Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

Section A **3*8=24**

1)	Determine the truth value of the statement: 'If it is raining, then the ground is wet' when it is not raining and the ground is still wet. Use propositional calculus to express this situation.	**3**	(CO1, L4)
2)	Prove that the statement 'All dogs are mammals' is valid using predicate calculus. Consider a domain where all elements are animals.	**3**	(CO1, L4)
4)	Express the statement 'There exists a number that is both even and prime' in predicate logic. Then, determine if it is true or false.	**3**	(CO1, L4)
5)	Formulate a valid argument using propositional logic and prove its validity through truth tables.	**3**	(CO1, L4)
6)	Using predicate calculus, reformulate the statement 'Every student in this class has read at least one book' and determine its validity.	**3**	(CO1, L4)
7)	Prove by induction that the sum of the first n odd numbers is equal to n^2.	**3**	(CO3, L4)
8)	Given a set of 5 elements, use the pigeonhole principle to prove that there must be at least two subsets with the same sum.	**3**	(CO4, L4)

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Section B — 8*7=56

1)	Prove that for any integer n, if n^2 is even, then n is also even.	**8**	(CO5, L4)
2)	Use the principle of mathematical induction to prove that for all n ≥ 1, the number (3n)! is divisible by 6^n.	**8**	(CO3, L4)
4)	Using the pigeonhole principle, prove that in any group of 10 people, at least two have their birthdays in the same month.	**8**	(CO4, L3)
5)	Prove that for any positive integer n, (n!)^2 > n^n.	**8**	(CO4, L3)

Section C — 10*2=20

1)	Prove by strong induction that for all integers k ≥ 2, the number (k^k) is greater than k!.	**10**	(CO3, L4)
2)	Using the principle of mathematical induction, prove that for all integers n ≥ 2, the sum of the first n natural numbers is equal to (n(n+1))/2.	**10**	(CO3, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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